Contents

1 On the correct choice of models

Polyhedra enjoy many nice visualization properties due to their 'small' nature: The picture most people -among those, the author- have in mind when talking about Polyhedra is a compact, connected subset of $\mathbb{R}^n$.

While this picture is amazing for reasoning and for applications, it gets in the way of the goal of this paper:

We will introduce Polyhedra, first in the 'concrete' setting, then in the abstract one, explaining what we believe to be the correct generalizations along the way.

1.1 Simplicial complexes

Simplicial complexes are the most common and useful way to describe polyhedra in the compact/finite case (i.e. the only one used in practice).

First a **Geometric description**

Definition 1.1 ((Geometric) simplicial complex)**.**

Let V be a set of points in $\mathbb{R}^n$. A ***Simplex*** *is the set of affine combinations*

$$\left\{\sum_i a_i v_i : a_i \in \mathbb{R}, \sum_i a_i = 1\right\}$$

A ***Simplicial Complex*** *or* ***Polyhedron*** *is a finite union of simplices.*

Note that any simplex is completely determined by its set V of vertices, allowing for a more compact (combinatorial) description

Definition 1.2 ((Combinatorial) simplicial complex)**.**

A ***Simplicial complex*** *is a (discrete) set $V \subseteq \mathbb{R}^n$ of* ***vertices*** *together with a set $\Sigma \subseteq \mathscr{P} V$ of* ***simplices*** *such that*

1. *Every face of a simplex is itself a simplex*
2. *Every intersection between two simplices is either empty or a common face of the two.*

We will often omit the set of vertices when talking about combinatorial simplicial complexes.

Given a polyhedron X we say that the simplicial complex Σ whose vertices and simplices are contained X is a ***triangulation*** *or* ***combinatorial reduction*** *of X, and conversely that X is a (the)* ***geometric realization*** *of Σ, denoted $|\Sigma|$.*

Lemma 1.3.

Every finite simplicial complex is the triangulation of a polyhedron.

Proof. □

Note that -contrary to what is usually thought of when we say 'face'- in both descriptions the vertices themselves are simplices, thus faces of the complex.

We need an appropriate notion of 'sameness' and of 'function' between two such objects, thus we need to define what are the components of the category we will be working in:

Note that **Top** (the category of topological spaces and continuous maps) is too loose for our benefit: A triangle and a square are homeomorphic -thus essentially the same objects as far as topology is concerned- but they're different objects to us. Differentiable manifolds are a bit too strict as the objects we deal with almost always have singular points;

The common solution in the literature is to consider **Piecewise-linear** manifolds and homeomorphisms between them.

Definition 1.4 (**PL**).

*The category **PL** of **Piecewise-linear manifolds** is the category that has as objects topological manifolds with an atlas whose transition maps are piecewise linear, and as maps continuous maps that are Piecewise-linear when restricted to the charts*

*In particular we call the category **PL**$_C$ of **Combinatorial manifolds** the full subcategory of **PL** whose objects are geometric simplicial complexes.*

Lastly before connecting the concepts we need to define what a morphism of simplicial complexes is and how to refine them (as a priori we do not have access to every point) of their faces.

Definition 1.5 (Morphism of (combinatorial) simplicial complexes). *A **morphism of simplicial complexes** $f : (V, \Sigma) \to (V', \Sigma')$ is a function $f : V \to V'$ such that the image of a simplex in Σ is a simplex of Σ'*

Theorem 1.6.

Let $f : X \to Y$ be a PL-morphism, then there exists a triangulation of X and Y such that f is a morphism of simplicial complexes.

Corollary 1.7.

For every pair of PL-homeomorphic geometric simplicial complexes $P \simeq P'$ there are two triangulations Σ of P and Σ' of P' that are isomorphic as combinatorial simplicial complexes.

Proof. □

Lastly, we want a way to speak about triangulations of simplicial complexes from a purely combinatorial point of view: that is, given a combinatorial simplicial complex, what other simplicial complexes realize to the same space?

Definition 1.8 (Stellar moves).

Given two disjoint simplicial complexes Σ and Σ' we define:
*Their **join** $\Sigma \vee \sigma'$ to be*

$$\{\sigma \cup \sigma' : \sigma \in \Sigma, \sigma' \in \Sigma'\}$$

*The **Star** of a face σ to be*

$$\bigstar_\Sigma(\sigma) := \{\tau \in \Sigma : \tau \cup \sigma \in \Sigma\}$$

*The **link** of a face σ to be*

$$\ell_\Sigma(\sigma) := \{\tau \in \bigstar_\Sigma(\sigma) : \tau \cap \sigma = \varnothing\}$$

*Lastly, the **deletion** of a face σ as*

$$\Sigma \setminus \sigma := \{\tau \in \Sigma : \sigma \nsubseteq \tau\}$$

*and the **boundary** of a face σ as*

$$\partial\sigma = \{\tau \in \Sigma : \tau \subsetneq \sigma\}$$

*A **stellar subdivision** of a complex Σ at σ is a simplicial complex*

$$(\Sigma \setminus \sigma) \cup (\{x\} \vee \partial\sigma \vee \ell_\Sigma(\sigma))$$

where x is a new vertex we are introducing.

*We call a **stellar move** of a complex either a stellar subdivision or its inverse, and we say that two complexes are **stellarly equivalent** $\Sigma \sim_\star \Sigma'$ if one can be obtained from the other via stellar moves.*

Theorem 1.9 (Alexander).

*Two simplicial complexes are stellarly equivalent if and only if their geometric realizations are **PL**-homeomorphic.*

Proof. □

All of this shows that we have an equivalence of categories between combinatorial manifolds and simplicial complexes modulo stellar equivalence.

1.2 Abstract Simplicial complexes

Recall that, as for Goldblatt-Thomason theorem, that a class of frames is modally definable if and only if

- It's closed under bounded morphic images,
- It's closed under (arbitrary) disjoint unions,
- It's closed under - and reflects ultrafilter extensions.

Images do nothing to subtract from our understanding of Polyhedra, but that is the only non-issue:

A class closed under arbitrary disjoint unions must contain a a non-compact non-connected space: Say we tried to generate a class from the polyhedron consisting of a single vertex, we can obtain an infinite discrete set of $\aleph_0$ vertices by taking the disjoint union of the point with itself $\aleph_0$ times.

An even bigger problem is the closure under ultrafilter extensions: Say we have a polyhedron with infinite cardinality $\aleph_0$ -which we must, as was argued just above-, its ultrafilter extension must have cardinality $2^{2^{\aleph_0}}$, which is too big to be a subset of $\mathbb{R}^n$ for any finite n.

Annoyingly enough, we expect those extension to be well-behaved dimension-wise (the ultrafilter extension an n-dimensional polyhedron should be n-dimensional too as they express the same logic) meaning that embedding it in $\mathbb{R}^\kappa$ for an appropriate cardinal κ wouldn't do us any favors, as we should always be able to find a finite-dimensional space (thus of smaller cardinality than our set) in which it's embeddable.

I believe we motivated enough in the previous section that the combinatorial side provides an adequate counterpart to the geometric side.

Definition 1.10 (Abstract simplicial complex).

*An **abstract simplicial complex** is a family of sets Σ closed under taking subsets. We rename the subset relation **face relation** and denote it with ≤ (and its irreflexive counterpart with <).*

Note that every notion defined in 1.7 makes sense in this context, thus we can finally define

Definition 1.11 (Abstract Polyhedron).

*A(n abstract) **Polyhedron** is an equivalence class of abstract simplicial complexes under stellar equivalence.*

Proposition 1.12 (Finite abstract equals combinatorial).

An abstract simplicial complex is (equivalent to) a combinatorial simplicial complex -in the sense of definition 1.2- if and only if it's finite. In other words there is an equivalence between the category of finite abstract simplicial complexes and combinatorial simplicial complexes.

Proof. If (V, Σ) is a combinatorial simplicial complex then Σ is a collection of sets closed under taking subsets, thus an abstract simplicial complex.

Now suppose we have a finite abstract simplicial complex Σ. We can construct V as the set of singletons of Σ. Every face σ can be expressed as a disjoint union of its elements, which must be elements of V as Σ is closed under taking subsets, thus without loss of generality $\sigma \in \mathscr{P} V$. Closure under subsets implies the two condition of definition 1.2. $\square$

From here on out we will omit the word "abstract" and instead refer to the content of the previous subsection as "finite" or "compact"

We want to check that these constructions have the properties we look for.

Recall that

Definition 1.13.

If (X, R) is a frame, then the ultrafilter extension $(\textbf{\textit{Uf}}(X), \tilde{R})$ is the frame whose underlying set is

$$\begin{aligned} \textbf{\textit{Uf}}(X) = \{U \subseteq \mathscr{P} X : & X \in U, \\ & \forall A \in U, \uparrow A \subseteq U, \\ & A, B \in U \Rightarrow A \cap B \in U \\ & \forall A \subseteq X, A \in U \text{ or } X \smallsetminus A \in U\} \end{aligned}$$

And the relation is

$$U \tilde{R} V \iff \{x \in X : \{y \in X : xRy\} \in V\} \in U$$

For the face relation (or any other order relation), this becomes

$$U \tilde{\prec} V \iff \{x \in X : \uparrow^{\prec} (x) \in V\} \in U$$

As you may notice, applying the ultrafilter extension to a simplicial complex does not result in a simplicial complex:

The basic reason is that no proper subset of an ultrafilter is an ultrafilter (for maximality), a deeper reason is that the construction takes into account only the face poset and not the set-theoretical content of the faces.

In other words, **Uf** takes in a poset and gives out a poset, thus we still need to realize it as a simplicial complex.

Definition 1.14 (Nerve of a poset)**.**

*Let $(P, \leq)$ be a poset. The **Nerve** $\mathscr{N}(P)$ is a simplicial complex constructed in the following way:*

- *The **vertices** are the points of P.*
- *The **sides** are the 1-chains $p_0 \leq p_1$ in P.*
- *The **Triangles** are the $2-$chains $p_0 \leq p_1 \leq p_2$ in P.*
- $\vdots$
- *The **n-simplices** are the $n-$chains $p_0 \leq p_1 \leq \cdots \leq p_n$ in P.*
- $\vdots$

This is clearly a simplicial complex as for any $m < n$, any sub-m-chain on an $n-$chain is part of the complex.

We will need the following theorem

Theorem 1.15 (Nerves behave well)**.**

Let Σ be a simplicial complex regarded as a poset with the face relation, then $\mathscr{N}(\Sigma)$ is stellarly equivalent to Σ.

Proof.

It's proven that the Nerve of the face poset of a simplicial complex gives a **barycentric subdivision** of the complex

and that any barycentric subdivision can be obtained through a sequence of stellar moves. □

We will now abuse notation a little bit and call the ultrafilter extension of a simplicial complex $\textbf{Uf}(\Sigma) := \mathscr{N}(\textbf{Uf}(\Sigma))$ the Nerve of the Ultrafilter extension of its face poset.

Proposition 1.16 (Closure under G-T)**.**

The class of abstract polyhedra is closed under ultrafilter extensions, disjoint unions and homomorphic images.

Proof.

Homomorphic images are immediate: Let $f : P \to Q$ be a morphism of polyhedra, i.e. an equivalence class of compatible morphisms of simplicial complexes. The image of any simplicial complex is -by definition- a simplicial complex and the compatibility conditions mean that all images are equivalent.

we have to show that disjoint unions and Ultrafilter extensions of equivalent simplicial complexes are equivalent.

Moreover, the ultrafilter extension of any finite preorder is isomorphic to the preorder itself, Thus its nerve falls in its equivalence class. so we can restrict to only looking at infinite simplicial complexes.

Suppose $\Sigma \sim_\star \Sigma'$ through a single stellar move on a face, this means that their ultrafilter extensions will be the same outside of that face meaning we can restrict to looking the local behaviour around the face, thus on a simplex.

Since the simplex is finite-dimensional $\textbf{Uf}(\Sigma) \simeq \Sigma \sim_\star \Sigma' \simeq \textbf{Uf}(\Sigma')$

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□

1.3 Polyhedral Models

Definition 1.17 (Polyhedral model)**.**

A ***Simplicial model*** *is a simplicial complex $K = (V, \Sigma)$ equipped with a valuation function $\nu :$* ***Prop*** *$\to \mathscr{P}\Sigma$*

A ***Polyhdral model*** *is a Polyhedron - in the sense of the previous section- $P = (V/\sim_p, \Sigma/\sim_p)$ equipped with a valuation. In other words a polyhedral model is a collection of simplicial models with a partially defined valuation that is compatible for all finer triangulations.*

In practice, when dealing with a polyhedral model, we fix a simplicial model that realizes the evaluation fully. Such a model will always exist for reasons of size.

We interpret the logic the usual way:

- $M, \sigma \vDash \top \iff$ always
- $M, \sigma \vDash \phi \iff \sigma \in \nu(\phi)$

- $M, \sigma \models \neg\phi \iff M, \sigma \not\models \phi$
- $M, \sigma \models \phi \wedge \psi \iff M, \sigma \models \phi$ and $M, \sigma \models \psi$
- $M, \sigma \models \Box\phi \iff \forall \tau \in \Sigma, \sigma \prec \tau \implies M, \tau \models \phi$

and all the other connectives can be derived from these.

Question: is there a purely combinatorial way to characterize when two simplicial complexes realize to the 'same' polyhedron?

YES Every two PL homeomorphic simplicial complexes have a common stellar subdivision

vattelo a pescare e produci (godo)

2 Goldblatt-Thomason Theorem